

1983

PERFORMANCE MEASUREMENT

THE "T P - 1" PERFORMANCE MODEL

Overview

STRATUS Performance

STRATUS vs. TANDEM

Observations

Future plans

STRATUS
INTERNAL USE ONLY

OVERVIEW
THE "TP-1" PERFORMANCE MODEL

Requester/Server model.

Requester - COBOL/TPF

Server - PL/1

Transactions initiated by delay interval. No screen I/o
or communications.

Servers are duplicated to achieve optimum performance
on each system.

Various file sizes and types.

Input parameters specify:

The number of reads and the number of updates for each
transaction.

Requester and server loop counts which are used to simulate
the processing workload of real transactions.

The number of server queues

File I/O designed to simulate random user access.

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File parameters

Small relative file	(20000 records, 150 bytes each)
Large relative file	(100000 records, 150 bytes each)
Large index file	(100000 records, 150 bytes each 12 byte key, no duplicates)

Each server was driven by a different random sequence read from a file at the beginning of the test.

Different random files were used for consecutive tests to minimize caching.

Static Static Use Only

TP - 1 OVERVIEW (continued)

Common parameters used for this comparison:

Number of terminals/tasks = 8

Delay interval = 2 seconds

Read count = 3

Update count = 2

Requester loop count = 100

Server loop count = 200

Number of requesters = 1

Number of servers = 2

States terminal use only

STRATUS PERFORMANCE

Small relative file (20000 records - 150 bytes each)

Non-duplexed system with 8MB memory

Non-duplexed 143MB disk

Results:

Transaction rate = 2.9 trans/sec

CPU utilization = 69%

Requester CPU utilization = 10%

Disk I/O rate = 18.0/sec

Observations:

Multi-tasking requester used relatively small
amount of cpu.

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STRATUS PERFORMANCE (continued)

Large relative file (100000 records, 150 bytes each)

Non-duplexed system with 8MB memory

Non-duplexed 143MB disk

Results:

Transaction rate = 2.3 trans/sec

CPU utilization = 94.7 %

Disk I/O rate = 21/sec

Observations:

Less cache benefits compared to small relative file

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STRATUS PERFORMANCE

Large indexed file

Non-duplexed system with either 4 or 8MB memory

Non-duplexed 143MB disk

Results:

8MB / 1 server queue / 1 disk:

Transaction rate = 1.5 trans/sec.

CPU utilization = 81 %

Disk I/O rate = 17/sec.

Requester CPU = .06 sec/trans.

Server CPU = .33 sec/trans.

8MB / 2 server queues / 1 disk:

Transaction rate = 1.6 trans/sec.

CPU utilization = 78 %

Disk I/O rate = 17/sec.

Requester CPU = .04 sec/trans.

Server CPU = .33 sec/trans.

4MB / 2 server queues / 1 disk:

Transaction rate = 1.5 trans/sec.

CPU utilization = 79 %

Disk I/O rate = 18/sec

Requester CPU = .04 sec/trans.

Server CPU = .34 sec/trans.

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STRATUS PERFORMANCE (continued)

Large indexed file (continued)

4MB / 2 server queues / 2 disks:

Transaction rate = 1.8 trans/sec

CPU utilization = 94 %

Disk I/O rate = 23/sec

Server CPU = .35 sec/trans.

4MB / 2 server queues / 4 servers / 2 disks:

Transaction rate = 1.7 trans/sec

CPU utilization = 98 %

Disk I/O rate = 22/sec

Server CPU = .35 sec/trans.

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STRATUS PERFORMANCE

- o Overhead in the multi-tasking requestor was minimal.
- o Task metering had no measurable effect.
- o Cache utilization significantly impacts performance. This is obviously application dependent, but significant improvements in performance are possible.
- o Stratus performance was most affected by disk issues - the type and number of disk accesses and the size of disk files.
- o Performance improved by adding a second disk drive.
- o Processor resource was saturated with two disks doing heavy indexed i/o.
- o Small improvement gain by using a second server queue.
- o Multiple server copies can greatly improve performance.
- o Process priorities are very important. Server priority should be higher than requestor. A production application should normally run at higher priority than development. Actual priorities may not be obvious because of the scheduler algorithms.
- o Preliminary multi-module recommendations:
 - Use Stratalink for queueing rather than file access.
 - Put queues on the requestor module.

Stratus Link used only

TANDEM PERFORMANCE TESTS

- o Used identical TP-1 model.
 - Same transaction workload.
 - Same data file.
 - Same record numbers and keys for random access.
- o PATHWAY requestor with TAL server.
- o Checkpointing done at system level (disk i/o) and in Pathway (interpreter and monitor processes).
- o Tests used single Tandem cpu.
- o Configuration for relative file tests:
 - TNSI cpu with 768KB memory
 - mirrored 64MB winchester disks
- o Configuration for indexed file tests:
 - TNSII cpu with 2MB memory
 - single 540MB winchester disk

States Actual Use Only

TEST RESULTS USING RELATIVE FILE (20,000 RECORDS)

TANDEM:

Transactions/sec	1.4
Requester cpu time/transaction	.37
Server cpu time/transaction (including disk processing)	.18
Disk I/O rate (physical)	15.3

STRATUS:

Transactions/sec	2.9
Requester cpu time/transaction	.03
Server cpu time/transaction (including disk processing)	.15
Disk I/O rate (physical)	18.0

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TEST RESULTS USING INDEXED FILE (100,000 RECORDS)

TANDEM:

Instructions in requestor loop

50 100 200

Transactions/sec	1.5	1.4	1.1
Requester cpu time/transaction	.31	.39	.61
Server cpu time/transaction (including disk processing)	.20	.20	.20
Disk I/O rate (physical)	22.2	22.0	17.5

STRATUS:

One Disk

Two Disks

Transactions/sec	1.5	1.8
Requester cpu time/transaction	.04	.04
Server cpu time/transaction (including disk processing)	.34	.35
Disk I/O rate (physical)	18.2	22.6

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STRATUS/TANDEM PERFORMANCE COMPARISON

- o Tandem disk drives are faster than current Stratus winchesters. *x20% faster*
- o Comparison looked only at single Tandem cpu and single Stratus module.
- o Tandem performance was most sensitive to instruction workload of the requestor (Pathway TCP). This is due to the fact that it interprets Screen Cobol and that it is checkpointing.
- o Stratus performance was most sensitive to the disk workload of the server.
- o The Tandem indexed file structure is more efficient than that of Stratus (at the expense of disk space) when accessing by primary key. When accessing by alternate key Tandem essentially doubles the time required to retrieve a record, while there is no difference between primary and alternate key access on Stratus. We did not have enough time to include alternate key access in our benchmark as yet.
- o Tandem offers very little flexibility in design options with their Pathway product which eliminates many opportunities for performance improvements. Examples of this include very limited data storage, no data sharing between tasks, and the inability to do any disk access in the requestor.
- o Tandem currently does not buffer disk writes in cache; writes go immediately to disk.
- o Tandem does no optimization for sequential disk i/o and performance in this area is very poor compared to Stratus.

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FUTURE PLANS

- o Multi-module
- o Stratanet
- o Component (atomic) testing
- o Communications products
- o System measurement and tuning
- o Transaction processing
- o Disk verify
- o Duplex versus simplex disks
- o File types and sizes
- o File locking types
- o New Stratus products
- o Competitive products when feasible

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